22. QUESTION D1: FORMATION OF THE EMBRYONIC PEDICLE

DURATION : 02:33

STUDY SHEET

COURSE SUMMARY

In this captivating video, we explore the formation of the embryonic stalk, a crucial moment in embryogenesis. We discover the dynamics of differential growth between the periphery and the centre of the embryo, and how this tearing movement contributes to the orientation and structuring of the embryonic stalk. The emergence of the latter is linked to mechanisms such as the growth of the analysis and the external coelom, influencing the creation of internal spaces. Finally, we address the transition from the embryonic stalk to the umbilical cord, highlighting the importance of the yolk sac in this process. This essential lesson offers an enriching perspective on biodynamic embryology and its clinical implications.

FORMATION OF THE EMBRYONIC PEDICLE

The establishment of the **third chamber** of the embryo is marked by a **differential growth rate**.

This rate is observed between the outer part of the embryo, which receives numerous trophic levels, and the centre, whose growth slows down compared to the outside. This phenomenon creates a movement, similar to a **tearing**, where the faster growth forms a layer of cells that appears on the edge.

THE CONCEPT OF "TEARING"

The term "tearing" is a **metaphor**. It is not a true tearing, but an image that evokes the dynamic movement within the embryo.

It is at the centre of the embryo that this movement occurs, and it is in this developmental process that the **embryonic pedicle** begins to form. Previously, it was uniformly distributed, but thanks to this movement, it orientates towards a central point.

ORIGIN OF THE EMBRYONIC PEDICLE

The phenomenon of embryonic pedicle formation is directly linked to the **differential growth rate**:

- Between the peripheral part of the embryo.
- Compared to the slowed rate of the centre.

This process occurs when the **astrocoele** begins to transform into a cavity. The ground is in a reaction position, which influences the growth system of the analysis. The latter, associated with the **growth of light**, also contributes to the formation of the body.

What is called the **growth of the analysis** and the **external coelom** ultimately gives rise to this space, creating a convergence of forces that leads to the formation of the embryonic pedicle.

FROM EMBRYONIC PEDICLE TO UMBILICAL CORD

It is important to note that the embryonic pedicle is not yet the **umbilical cord**. At a certain point, due to the excessive growth of the **amniotic cavity**, it will push the **vitelline vesicle** onto the embryonic pedicle.

The meeting between the embryonic pedicle and the vitelline vesicle is what will form the umbilical cord.